

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1151

Detour 138 kV
March 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1151. The Interconnection Customer proposes a 203.5 MW solar photovoltaic facility interconnecting at the Detour 138 kV substation in the Virginia area. The request was received on February 15, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$81,772,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TOA VACA / WOODS DP 138.0 kV	TOA VACA / WOODS DP 138.0 kV	124.8	331	8.4
PASO CNTY SOLID WASTE RESOURCE RECOVERY / NOVEC PARADISE 138.0 kV Line	SOUTH JORDAN / CHAFFEE GAS RECOVERY 138.0 kV	123.3	245	7.6
VMEA EAKING GEN / WOODS DP 138.0 kV	WEST STAUNTON SUBSTATION / WOODS DP 138.0 kV Line	118.8	234	5.3
FORT BENTON / MOUNT PLEASANT SUBSTATION 138.0 kV	FORT BENTON / MIDDLETOWN JCT 138.0 kV	116.6	229	30.5
FT BENTON / MIDDLETOWN JCT 138.0 kV	FT BENTON / MOUNT PLEASANT 138.0 kV	115.2	219	30.8
MIDDLETOWN JUNCTION / MOUNT PLEASANT 138.0 kV	FT BENTON / MOUNT PLEASANT 138.0 kV	114.9	220	18.9
CHAFFEE GAS RECOVERY / S JORDAN 138.0 kV	WHEELER / FIREBANTZ 138.0 kV L	114.0	306	14.8
SILAM SPRINGS / DETOUR SUBSTATION 138.0 kV Line	FT BENTON / DETOUR 138.0 kV	113.8	352	59.4
DETOUR / FT BENTON 138.0 kV	SILAM SPRINGS / DETOUR 138.0 kV	111.1	424	63.3
TOA VACA / WOODS DP 138.0 kV	TOA VACA / WOODS DP 138.0 kV	110.9	372	8.4
FIREBANTZ / MOUNT PLEASANT 138.0 kV	SOUTH JORDAN / CHAFFEE GAS RECOVERY 138.0 kV	110.7	307	22.0
S JORDAN / MOUNT PLEASANT 138.0 kV Line	FIREBANTZ / MOUNT PLEASANT 138.0 kV	104.9	232	23.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TAMPA BAY / LORIN INDUSTRIES 138.0 kV line	TAMPA BAY / DPC PRAIRIE DU CHIEN 138.0 kV Line	101.4	181	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WOODS DP / TOA VACA 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,136,000	22% (\$247,000)
PASCO CNTY SOLID WASTE RESOURCE RECOVERY SUBSTATION / NOVEC PARADISE 138.0 kV	Rebuild line with higher-capacity conductor	\$3,444,000	30% (\$1,046,000)
VMEA PEAKING GEN / WOODS DP 138.0 kV	Rebuild line with higher-capacity conductor	\$5,112,000	25% (\$1,277,000)
MOUNT PLEASANT / FORT BENTON 138.0 kV Line	Rebuild line with higher-capacity conductor	\$34,666,000	67% (\$23,230,000)
MIDDLETOWN JCT / FORT BENTON 138.0 kV	Rebuild line with higher-capacity conductor	\$25,026,000	67% (\$16,874,000)
MOUNT PLEASANT SUBSTATION / MIDDLETOWN JUNCTION 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,041,000	70% (\$3,523,000)
CHAFFEE GAS RECOVERY / S JORDAN 138.0 kV	Rebuild line with higher-capacity conductor	\$9,795,000	48% (\$4,753,000)
SILOAM SPRINGS / DETOUR SUBSTATION 138.0 kV Line	Rebuild line with higher-capacity conductor	\$26,120,000	68% (\$17,770,000)
FT BENTON / DETOUR 138.0 kV	Rebuild line with higher-capacity conductor	\$7,261,000	68% (\$4,918,000)
WOODS DP / TOA VACA 138.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,213,000	22% (\$263,000)
FIREBANTZ / MOUNT PLEASANT 138.0 kV	Rebuild line with higher-capacity conductor	\$9,198,000	50% (\$4,635,000)
S JORDAN / MOUNT PLEASANT 138.0 kV L	Reconductor line	\$4,260,000	71% (\$3,018,000)
TAMPA BAY / LORIN INDUSTRIES 138.0 kV line	Reconductor line	\$1,785,000	12% (\$218,000)
	Total allocated to project		\$81,772,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.